

ETLens AI

ETL Script Documentation Generator

Test Cases Document

1. File Upload & Parsing

TC#	Test Case	Input	Expected Output	Status
TC-01	Upload Python ETL script	sample_etl.py	File parsed, functions and classes extracted	✓ Pass
TC-02	Upload SQL ETL script	sample_etl.sql	SQL tables, joins, and transformations extracted	✓ Pass
TC-03	Upload unsupported file type	.txt file	Error message shown to user	✓ Pass
TC-04	Upload empty file	empty.py	Graceful error, no crash	✓ Pass
TC-05	Upload large Python script	large_etl.py (500+ lines)	Parsed successfully within timeout	✓ Pass

2. Documentation Generation

TC#	Test Case	Input	Expected Output	Status
TC-06	Generate docs from cache	Pre-cached sample_etl.py	Documentation returned instantly (<1s)	✓ Pass
TC-07	Generate docs for new file	New Python script	Template-based docs generated instantly	✓ Pass
TC-08	Generate SQL documentation	sample_etl.sql	Table descriptions and transformations documented	✓ Pass
TC-09	Documentation completeness	Any ETL script	Functions, params, return values all documented	✓ Pass
TC-10	LLM timeout fallback	New file, LLM slow	Template fallback triggered, no hang	✓ Pass

3. Business Explanation

TC#	Test Case	Input	Expected Output	Status
TC-1 1	Business explanation for Python	sample_etl.py	Plain-language business purpose generated	✓ Pass
TC-1 2	Business explanation for SQL	sample_etl.sql	Business context and data flow explained	✓ Pass
TC-1 3	Cache hit for business explanation	Pre-cached file	Explanation returned instantly from cache	✓ Pass

4. Flow Diagram Generation

TC#	Test Case	Input	Expected Output	Status
TC-1 4	Generate flow diagram	sample_etl.py	Graphviz PNG diagram showing data flow	✓ Pass
TC-1 5	SQL flow diagram	sample_etl.sql	Table dependency diagram generated	✓ Pass
TC-1 6	Diagram displays in UI	Any script	PNG renders correctly in Streamlit	✓ Pass

5. Impact Analysis

TC#	Test Case	Input	Expected Output	Status
TC-1 7	Generate impact analysis	sample_etl.py	NetworkX dependency graph generated	✓ Pass
TC-1 8	Identify table dependencies	sample_etl.sql	Upstream/downstream tables identified	✓ Pass
TC-1 9	Impact graph display	Any script	Dependency graph renders in Streamlit UI	✓ Pass

6. RAG-based Q&A

TC#	Test Case	Input	Expected Output	Status
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TC-2 0	Ask question about script	'What does this script do?'	Relevant answer based on script content	✓ Pass
TC-2 1	Ask about specific function	'What does extract_data() do?'	Function-specific answer returned	✓ Pass
TC-2 2	Ask about SQL tables	'Which tables are used?'	List of tables from SQL script returned	✓ Pass
TC-2 3	FAISS index creation	New script uploaded	Vector index created successfully	✓ Pass

7. PDF Export

TC#	Test Case	Input	Expected Output	Status
TC-2 4	Export documentation as PDF	Any documented script	PDF downloaded with all sections	✓ Pass
TC-2 5	PDF contains all sections	sample_etl.py	Docs, business explanation, flow diagram in PDF	✓ Pass
TC-2 6	PDF formatting	Any script	Clean, professional formatting in PDF	✓ Pass

8. API Endpoints

TC#	Endpoint	Method	Expected Response	Status
TC-2 7	/upload	POST	File parsed, 200 OK returned	✓ Pass
TC-2 8	/generate-doc	POST	Documentation JSON returned	✓ Pass
TC-2 9	/business-explain	POST	Business explanation returned	✓ Pass
TC-3 0	/flow-diagram	POST	PNG image returned	✓ Pass
TC-3 1	/impact-analysis	POST	Dependency graph data returned	✓ Pass
TC-3 2	/ask	POST	RAG-based answer returned	✓ Pass
TC-3 3	/export-pdf	POST	PDF file returned for download	✓ Pass

Test Summary

Category	Total Tests	Passed	Failed	Pass Rate
File Upload & Parsing	5	5	0	100%
Documentation Generation	5	5	0	100%
Business Explanation	3	3	0	100%
Flow Diagram Generation	3	3	0	100%
Impact Analysis	3	3	0	100%
RAG-based Q&A	4	4	0	100%
PDF Export	3	3	0	100%
API Endpoints	7	7	0	100%
TOTAL	33	33	0	100%